Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN-Shift-1 | Date: 21-03-2026

Time: 09.00Am to 12.00Pm | GTM-25 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**3**	2	**3**	3	**3**	4	**3**	5	**2**
6	**3**	7	**1**	8	**3**	9	**1**	10	**2**
11	**2**	12	**2**	13	**3**	14	**2**	15	**1**
16	**1**	17	**1**	18	**2**	19	**3**	20	**4**
21	**70**	22	**15**	23	**3**	24	**2**	25	**5**

PHYSICS

26	**1**	27	**1**	28	**1**	29	**3**	30	**1**
31	**1**	32	**3**	33	**3**	34	**4**	35	**2**
36	**2**	37	**4**	38	**4**	39	**4**	40	**2**
41	**2**	42	**2**	43	**2**	44	**3**	45	**3**
46	**18**	47	**5**	48	**208**	49	**15**	50	**4**

CHEMISTRY

51	**2**	52	**4**	53	**4**	54	**2**	55	**2**
56	**3**	57	**2**	58	**4**	59	**3**	60	**3**
61	**2**	62	**3**	63	**3**	64	**3**	65	**3**
66	**1**	67	**1**	68	**2**	69	**4**	70	**1**
71	**2**	72	**4**	73	**5**	74	**11**	75	**423**

For More Join: @IITJEE_Advanced

SOLUTIONS
MATHEMATICS

1. $S = 7 + 19 + 39 + 67 + \ldots\ldots\ldots + t_n$

$S = 7 + 19 + 39 + 67 + \ldots\ldots\ldots + t_{n-1} + t_n$

$O = 7 + 12 + 20 + 28 + \ldots\ldots\ldots - t_n$

$t_n = 7 + \frac{n-1}{2}\left(24 + (n-2)8\right)$

$t_n = 4n^2 + 3$

$S_n = \sum_1^n \tan^{-1}\left(\frac{4}{4n^2+3}\right)$

$= \sum_1^n \tan^{-1}\left(\frac{\left(n+\frac{1}{2}\right)-\left(n-\frac{1}{2}\right)}{1+\left(n+\frac{1}{2}\right)\left(n-\frac{1}{2}\right)}\right)$

$= \sum_1^n \tan^{-1}\left(n+\frac{1}{2}\right) - \tan^{-1}\left(n-\frac{1}{2}\right)$

$S_n = \tan^{-1}\left(n+\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{2}\right)$

$S_\infty = \frac{\pi}{2} - \tan^{-1}\left(\frac{1}{2}\right) = \cot^{-1}\left(\frac{1}{2}\right) = \tan^{-1} 2$

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2. Equation of a tangent to the parabola $y^2 = 4x$ is $y = mx + 1/m$. It will touch the circle $(x-3)^2 + y^2 = 9$ whose centre is $(3,0)$ and radius is 3 if

$\left|\frac{0 + m(3) + (1/m)}{\sqrt{1+m^2}}\right| = 3 \Rightarrow (3m + 1/m)^2 = 9(1+m^2)$

$\Rightarrow 9m^2 + 6 + 1/m^2 = 9 + 9m^2 \Rightarrow m^2 = 1/3 \Rightarrow m = \pm 1/\sqrt{3}$

As the tangent is above the x - axis, we take $m = 1/\sqrt{3}$ and thus the required equation is $\sqrt{3}y = x + 3$.

3. Let $y = \frac{ax^2 + 3x - 4}{a + 3x - 4x^2} \Rightarrow ay + 3yx - 4x^2 y = ax^2 + 3x - 4$

$\Rightarrow (a + 4y)x^2 + 3(1-y)x - (4 + ay) = 0$

Since x is real, $\Delta \geq 0 \Rightarrow 9(1-y)^2 + 4(a+4y)(4+ay) \geq 0$

$\Rightarrow 9(y^2 - 2y + 1) + 4(4a + a^2 y + 16y + 4ay^2) \geq 0$

$\Rightarrow 9y^2 - 18y + 9 + 16a + 4a^2 y + 64y + 16y^2 \geq 0$

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$\Rightarrow (9+16a)y^2+46y+4a^2y+(9+16a)\geq 0$

$ax^2+bx+c\geq 0\ \forall x\in R\Rightarrow \Delta\leq 0.$

$\therefore (46+4a^2)^2-4(9+16a)^2\leq 0\Rightarrow 4\left[\left(2a^2+23\right)^2-\left(9+16a\right)^2\right]\leq 0$

$\Rightarrow (2a^2+23+9+16a)(2a^2+23-9-16a)\leq 0\Rightarrow (2a^2+16a+32)(2a^2-16a+14)\leq 0$

$\Rightarrow 4(a^2+8a+16)(a^2-8a+7)\leq 0\Rightarrow (a+4)^2(a-1)<0\Rightarrow 1\leq a\leq 7.\quad \therefore a\in[1,7].$

4. Let $\alpha,\beta,\gamma,\delta$ are roots of

$x^4-ax^3+bx^2-cx+d=0$

Apply $A.M\geq G.M$

$\alpha,\beta,\gamma,\delta,\sum\alpha\beta,\sum\alpha\beta\gamma,$

$$\frac{\alpha+\beta+\gamma+\delta+\alpha\beta+\alpha\gamma+\alpha\delta+\beta\gamma+\beta\delta+\gamma\delta+\alpha\beta\gamma+\alpha\beta\delta+\alpha\gamma\delta+\beta\gamma\delta}{14}$$

$$\geq(\alpha.\beta.\gamma.\delta.\alpha\beta.\alpha\gamma.\alpha\delta.\beta\gamma.\beta\delta.\gamma\delta.\alpha\beta\gamma.\alpha\beta\delta.\alpha\gamma\delta.\beta\gamma\delta)^{1/14}$$

$$=\frac{\sum\alpha+\sum\alpha\beta+\sum\alpha\beta\gamma}{14}\geq(\alpha^7.\beta^7.\gamma^7.\delta^7)^{1/14}$$

$$\Rightarrow\frac{a+b+c}{14}\geq(\alpha\beta\gamma\delta)^{1/2}$$

5. Any point P on the given line is $(2r+1,-3r-1,8r-10)$

So the direction ratios of AP are $(2r,-3r-1,8r-10)$

Now AP is perpendicular to the given line

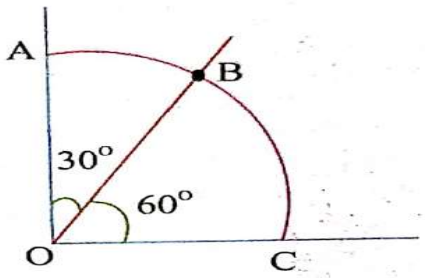
$\Rightarrow 2(2r)-3(-3r-1)+8(8r-10)=0\Rightarrow 77r-77=0\Rightarrow r=1$

And thus the coordinates of P, the foot of the perpendicular from A on the line are $(3,-4,-2)$.

Let $B(a.b,c)$ be the reflection of A in the given line. Then P is the mid – point of AB.

$$\frac{a+1}{2}=3,\frac{b}{2}=-4,\frac{c}{2}=-2\Rightarrow a=5,b=-8,c=-4\Rightarrow B=(5,-8,-4).$$

6.



Given that $\overline{OA},\overline{OB}$, *and* $\overline{OC}$ are coplanar, then $\overline{OB}$ can be written as the linear combination of two non – collinear vectors $\overline{OA}$ and $\overline{OC}$. Let $\overline{OC}$ and $\overline{OA}$ be taken along x and y axes respectively, and

$|\overline{OA}|=|\overline{OB}|=|\overline{OB}|=r$

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Where r be the radius, then we can write $\left|\overline{OB}\right| = r(\cos 60^o)\dfrac{\overline{OC}}{\left|\overline{OC}\right|} + r(\cos 30^o)\dfrac{\overline{OA}}{\left|\overline{OA}\right|}$

$\bar{b} = \dfrac{1}{2}\overline{OC} + \dfrac{\sqrt{3}}{2}\bar{a} \Rightarrow \overline{OC} = 2\bar{b} - \sqrt{3}\,\bar{a}$

7. $\int_0^{[x]} 2^{x-[x]} = [x]\int_0^1 2^{x-[x]}dx = [x]\int_0^1 2^x dx$

8. Given system of equation has nontrivial solution $\Rightarrow \begin{vmatrix} t & t+1 & t-1 \\ t+1 & t & t+2 \\ t-1 & t+2 & t \end{vmatrix} = 0$

$R_2 - R_1, R_3 - R_1 \begin{vmatrix} t & t+1 & t-1 \\ 1 & -1 & 3 \\ -1 & 1 & 1 \end{vmatrix} = 0 \Rightarrow t(-1-3) - (t+1)(1+3) + (t-1)(1-1) = 0$

$\Rightarrow -4t - 4(t+1) = 0 \Rightarrow -8t - 4 = 0 \Rightarrow t = -1/2.$

9. We have $Lf'(1) = \lim_{h\to 0} \dfrac{f(1-h) - f(1)}{-h}$

$= \lim_{h\to 0} \dfrac{f(1-h)[1-h] - 1[1]}{-h} = \lim_{h\to 0} \dfrac{0-1}{-h} \to \infty$

$f'(1)$ does not exist.

Also, $Lf'(2) = \lim_{h\to 0} \dfrac{f(2-h) - f(2)}{-h}$

$= \lim_{h\to 0} \dfrac{f(2-h)[2-h] - (2-1)[2]}{-h} = \lim_{h\to 0} \dfrac{2-h-1}{-h} = 1$

And, $Rf'(2) = \lim_{h\to 0} \dfrac{f(2+h) - f(2)}{h}$

$= \lim_{h\to 0} \dfrac{f(2+h)[2+h] - (2-1)[2]}{h} = \lim_{h\to 0} \dfrac{2(1+h)-2}{h} = \lim_{h\to 0} \dfrac{2h}{2} \to 2$

$f'(2)$ also does not exist.

10. Solving $x^2 + y^2 = 16a^2, y^2 = 6ax$

$x^2 + 6ax - 16a^2 = 0 \Rightarrow (x+8a)(x-2a) = 0$

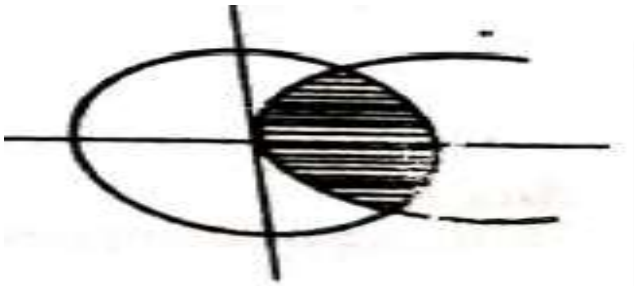
$\Rightarrow x = -8a, 2a$

$x - 2a$ *then*, $y^2 = 12a^2 \Rightarrow y = 2\sqrt{3}a,$

$$Area = 2\int_0^{2\sqrt{3}a} \left(\sqrt{16a^2 - y^2} - \frac{y^2}{6a}\right)dy = 2\left[\frac{y}{2}\sqrt{16a^2 - y^2} + 8a^2 \sin^{-1}\frac{y}{4a} - \frac{y^3}{18a}\right]_0^{2\sqrt{3}a}$$

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$$=2\left[\sqrt{3}a.2a+8a^2.\frac{\pi}{3}-\frac{4}{\sqrt{3}a^2}\right]=2\left[8a^2.\frac{\pi}{3}+\frac{2}{\sqrt{3}}a^2\right]=\frac{4a^2}{3}(4\pi+\sqrt{3})$$



11. $(1-x)^{-p/q}$

12. f is continuous at $x=0\Rightarrow \underset{x\to 0^+}{Lt} f(x)=\underset{x\to 0^-}{Lt} f(x)=f(0)$

$$=\underset{x\to 0^+}{Lt}\left(\alpha+\frac{\sin[x]}{x}\right)=\underset{x\to 0^-}{Lt}\left(\beta+\left[\frac{\sin x-x}{x^3}\right]\right)=2$$

$$\Rightarrow \alpha=\beta-1=2\Rightarrow \alpha=2,\ \beta=3\Rightarrow \beta-\alpha=1.$$

13. Put $y=\frac{1}{x}\to 2+f(x).f\left(\frac{1}{x}\right)$

$$=f(x)+f\left(\frac{1}{x}\right)+f(1)---(1)$$

Put $x=1$

$2+\left(f(1)\right)^2=3f(1)\to f(1)=1(or)2$

$f(1)\neq 1$, Otherwise, from the given relation

$2+f(x)f(1)=f(x)+f(1)+f(x)\ \ (or)$

$f(x)=1$, Which is not possible as $f(2)=5$

$f(1)=2\quad from\ (1)$

$f(x).f\left(\frac{1}{x}\right)=f(x)+f\left(\frac{1}{x}\right)$

$\to f(x)=\pm x^n+1\Rightarrow f\left(f(2)\right)=26.$

14. $i)I.F=\exp\left[\int\frac{x^2}{x^3+1}dx\right]=\exp\left[\frac{1}{3}\log(x^3+1)\right]=(x^3+1)^{1/3}\to(e)$

$ii)I.F=\exp\left[\int\frac{3x}{x^2}dx\right]=\exp(3\log x)=x^3\to(a)$

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$iii)\, I.F = \exp\left[\int \frac{6x^2(x^3+1)}{(x^3+1)^2}dx\right] = \exp\left[\int \frac{6x^2}{x^3+1}dx = \exp\left[2\log\left(x^3+1\right)\right]\right]$

$= (x^3+1)^2 \to (b)$

$iv) I.F = \exp\left[\int \frac{4x}{x^2+1}dx\right] = \exp\left[2\log(x^2+1)\right] = (x^2+1)^2 \to (c).$

15. $(a,b),(c,d) \in N \times N; (a,b)R(c,d) \Rightarrow ab(b+c) = bc(a+d)$

Reflexive: Since $ab(b+a) = ba(a+b) \forall ab \in N.$ $\therefore (a,b)R(a,b)$ $\therefore R$ is reflexive.

Symmetric: For $(a,b)R(c,d) \in N \times N$, $Let (a,b)R(c,d).$

$\therefore ad(b+c) = bc(a+d) \Rightarrow bc(a+d) = ad(b+c)$

$\Rightarrow cb(d+a) = da(c+b) \Rightarrow (c,d)R(a,b).$

$\therefore R$ is symmetric.

Transitive: For $(a,b),(c,d)(e,f) \in N \times N$, $Let\ (a,b)R(c,d),(c,d)R(e,f)$

$\therefore ad(b+c) = bc(a+d), cf(d+e) = de(c+f)$

$\Rightarrow adb + adc = bca + bcd \to (1)$

$cfd + cfe + = dec + def \to (2)$

$(1)ef + (2)ab \Rightarrow adbef + adcef + cfeab$

$= bcaef + bcdef + decab + defab$

$\Rightarrow adcf(b+e) = bcde(a+f) \Rightarrow af(b+e) = be(a+f) \Rightarrow (a,b)R(e,f).$

$\therefore R$ is transitive. Hence R is an equivalence relation.

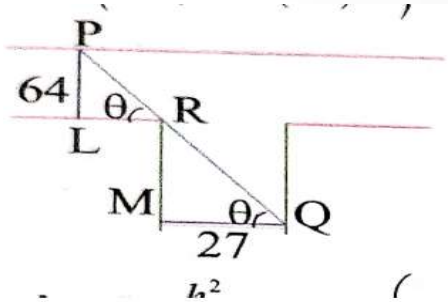
16. Put $ax + \frac{b}{x} = t$

17. $\overline{OB} = \left|\overline{OB}\right| e^{i\theta}$

$\left|\overline{OB}\right| = |OA| = \sqrt{3+1} = \sqrt{4} = 2$

$\overline{OB} = \left|\overline{OB}\right| e^{i\frac{\pi}{2}} = 2.e^{i\pi/2} = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

18.



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$$\sin\theta = \frac{64}{PR} \Rightarrow PR = 64\cos ec\theta$$

$$\cos\theta = \frac{27}{RQ} \Rightarrow RQ = 27\sec\theta$$

$$PQ = PR + RQ = 64\cos ec\theta + 27\sec\theta$$

Min Value of PQ is $(a^{2/3} + b^{2/3})^{3/2} = \left((27)^{2/3} + (64)^{2/3}\right)^{3/2} = 125$

19. $$P(A) = \frac{{}^{2}C_1 \times {}^{32}C_2 \times {}^{32}C_1}{{}^{64}C_3}$$

20. $$\sum_{r=0}^{n}(-1)^r\,{}^nC_r\left[i^{5r} + i^{6r} + i^{7r} + i^{8r}\right] = \sum_{r=0}^{n}(-1)^r\,{}^nC_r\left[i^r + i^{2r} + i^{3r} + 1\right]$$

$$= \sum_{r=0}^{n}(-1)^r\,{}^nC_r\,i^r + \sum_{r=0}^{n}(-1)^r\,{}^nC_r\left(i^2\right)^r + \sum_{r=0}^{n}(-1)^r\,{}^nC_r\left(i^3\right)^r + \sum_{r=0}^{n}(-1)^r\,{}^nC_r$$

$$= (1-i)^n + (1-i^2)^n + (1-i^3)^n + (1-1)^n = (1-i)^n + 2^n + (1+i)^n$$

$$= 2^n + 2^{\pi/2}\left\{\cos\frac{\pi}{4} - i\sin\frac{\pi}{4}\right\}^n + 2^{\pi/2}\left\{\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right\}^n = 2^n + 2^{\frac{n}{2}+1}\cos\frac{n\pi}{4}.$$

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21. $$4m = 2^a + 3^b + 5^c = 2^a + (4-1)^b + (4+1)^c$$

$$\Rightarrow 4m = 4k + 2^a + (-1)^b + (1)^c$$

$$\Rightarrow a = 1, b = even, c = any\ number\ or\ a \neq 1, b = odd, c = any\ number$$

$$\therefore \text{Re}quired\ number = 1\times2\times5 + 4\times3\times5 = 70.$$

22. $$\text{Median} = \frac{420+420}{2} = 420.$$

Mean Deviation=

$$\frac{\begin{array}{l}|390-420| + |400-420| + |400-420| + |410-420| + |410-420| + |420-420| \\ + |420-420| + |430-420| + |430-420| + |440-420| + |440-420| + |450-420|\end{array}}{12}$$

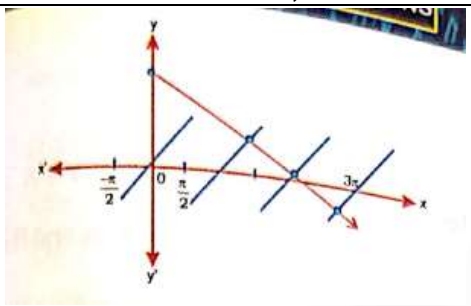
$$= \frac{30+20+20+10+10+0+0+10+10+20+20+30}{12} = \frac{180}{12} = 15.$$

23. $$\tan^{-1}(\tan x) = 3 - \frac{x}{2}$$

Draw the graph of $y = \tan^{-1}(\tan x)$

$y = 3 - x/2$

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24. A and B are on opposite sides of y - axis. Reflection of $A(-3,2)$ in y – axis is $A'=(3,2)$. So that A', B
Will be on the same side of P Equation of $A'B$ is
$y-2=-3(x-3)\Rightarrow y-2=-3x+9$
$3x+y=11$
The meets y – axis is at $Q(0,11)$
Q is required position of P so that $k=11$
$\therefore \frac{k-1}{5}=\frac{10}{5}=2$
(Since difference of two sides of a triangle is smaller than the third side)
Note: We know that $|PA-PB|$ will be greatest if P lies on the line $\overleftrightarrow{AB}$, and A and B must lie on same side of P.

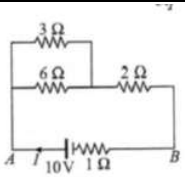
25. Area of quadrilateral $ABCD = Area\ of\ \Delta ABC+\Delta ACD$
$=\frac{1}{2}\left[a\times(2a+3b)\right]+\frac{1}{2}\left[\left[2a+3b\right]\times b\right]=\left[3a\times b+2a\times b\right]=\frac{5}{2}a\times b=\frac{5}{2}$
Area of parallelogram $ABCD$.
$\therefore \alpha=\frac{5}{2}$

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PHYSICS

26. Both A and R are correct and R is the correct explanation of A
27. Both A and R are true and R is the correct explanation of A
28. Statement I is correct but statement II is incorrect.
29. Statement-I & Statement-II are true.
30. When $P = mg(\sin\theta - \mu\cos\theta)$ $f = \mu mg\cos\theta(upwards)$
When $P = mg\sin\theta$ $f = 0$
And when $P = mg(\sin\theta + \mu\cos\theta)$ $f = \mu mg\cos\theta(downwards)$
Hence friction is first positive, then zero and then negative.b
31. Acceleration =v $\frac{dv}{dx}$
32. $n\ MSD = (n+2)VSD \quad L.C = 1MSD - 1VSD$
33. Since velocity increase from zero to positive value so particle starts from $y = -A$ towards mean position.
34. Let us plot the graph of the mathematical equation
$U(x) = K\left[1 - e^{-x^2}\right] \quad \therefore \quad F = -\frac{dU}{dx} = -2kxe^{-x^2}$
It is clear that the potential energy is minimum at $x = 0$. Therefore, $x = 0$ is the state of stable equilibrium. Now if we displace the particle from $x = 0$, then for small displacements the particle tends to regain the position $x = 0$ with a force $F = \frac{2kx}{e^{x^2}}$ for x To be small $F\ \alpha\ x$.
35. Loss of Heat = $\frac{dQ}{dt}\alpha AT^4$
36. $M\ \alpha\ R^3 \quad (\because M = V \times p)$ Mass of cavity $\rightarrow M/8$
$F = \frac{GMm}{R^3}\left(\frac{R}{2}\right) - \frac{G\left(\frac{M}{8}\right)m}{R^2} = \frac{GMm}{R^2}\left[\frac{4}{8} - \frac{1}{8}\right] = \frac{3GMm}{8R^2}$
$\therefore F = \frac{3}{8}mg \quad (\because g = GM/R^2)$
37. $r^{1/2}$
38. $\frac{\Delta V}{V} = -(1-2\sigma)\frac{\Delta l}{l} \quad \frac{\Delta V}{V} \times 100 = -[1-2(0.3)]4\times10^3\times100$
$= -(1-0.6)4\times10^{-3}\times100 \quad = -0.4\times4\times10^{-3}\times100 \quad = -1.6\times10^{-1} = -0.16\%$
So, volume decreases by 0.16%.
39. Potential difference independent of charge on outer sphere.
40. $R_{eq} = \frac{3\times6}{3+6} + 2 + 1 = 5\Omega\ I = \frac{10}{R_{eq}} = \frac{10}{5} = 2A\ V_A - V_B = 10 - I\times1 = 8V$

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$$\frac{V_A - V_B}{10} = 0.8.$$

41. For Secondary minima, $b\sin\theta = n\lambda \Rightarrow \sin\theta = \frac{n\lambda}{b}$

Distance of nth secondary minima $x = D\sin\theta$ Or

$\sin\theta_1 = \frac{x_1}{D} \Rightarrow \sin\theta_1 = \frac{2\lambda}{b}$ $n = 4$

$\sin\theta_2 = \frac{4\lambda}{b} = \frac{x_2}{D}$ $\frac{x_2 - x_1}{D} = \frac{4\lambda}{b} - \frac{2\lambda}{b} = \frac{2\lambda}{b}$

Width of central maxima $= D\frac{2\lambda}{b} = 3cm.$

42. $\vec{E} \times \vec{B}$ gives direction of $\vec{V}$.

43. $M = \frac{\beta}{\alpha} = \frac{f_0}{f_e}$ $\therefore \beta = \frac{f_0}{f_e}\alpha = \frac{60}{5} \times 2^0 = 24^0$

44. $E_n = -13.6\frac{Z^2}{n^2}ev,\ r_n = 0.529\frac{n^2}{z}A^0,\ V_n = 2.2 \times 10^6 \frac{z}{n} m/s,\ T = \frac{2\pi r_n}{v_n}$

45. $Y = \overline{\overline{(\overline{A} + \overline{B})}} = \overline{A.B}$ This is NAND gate.

46. $rF = I\alpha$ $\alpha = 4t - t^2$ $w = 2t^2 - t^{3}/_{3}$ $w = 0$ $t = 6$ $\theta = 2t\ ^{3}/_{3} - t\ ^{4}/_{12}$

47. The ratio $\frac{P_2}{P_1}$ is 5

48. $^{235}U \rightarrow ^{140}Ce + ^{94}Zr + n$ Disintegration energy

$Q = \left(m_R - m_P\right).c^2,\ m_R = 235.0439\ u$

$m_P = 139.9054u + 93.9063u + 1.0086u = 234.8203u$

$Q = \left(235.0439u - 234.8203u\right)c^2 = 0.2236c^2 = 0.2236 \times 931$ $Q = 208.1716.$

49. $KE_{\max} = \frac{hc}{\lambda} - \Phi$

50. $x_1 = 1\omega = 2\pi fl = 3.14\Omega$ Also, $\tan\Phi = \frac{x_L}{R}$

$$= \frac{\tan^{-1}\left(\frac{\omega L}{R}\right)}{\omega} = \frac{\tan^{-1}\left(\frac{Z}{R}\right)}{\omega} = \frac{\tan^{-1}(3.14)}{\omega} = \frac{72^o}{\omega} = \frac{72 \times \pi}{180} = \frac{1}{250}S$$

$$\frac{2\pi \times 50}{}$$

$4 \times 10^{-3} S = k$ So, $1000k = 4.$

For More Join: @IITJEE_Advanced

CHEMISTRY

51. O_2^{+1}, O_2^{-1} both have one electron in π^* molecular orbital.

52. For the given reaction,

$$CH_3OH_{(l)} + \frac{3}{2}O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}$$

$$\Delta G^o_{reaction} = \sum \Delta G^o_{f(products)} - \Delta G^o_{f(\text{reactants})}$$

$$= \left[-394.4 + 2\times(-237.2)\right] - (-166.2) = -702.6\ kJ\ mol^{-1}$$

$$\%\text{Efficiency} = \frac{\Delta G}{\Delta H}\times 100 = \frac{-702.6}{-726}\times 100 = 96.77\% \approx 97\%.$$

53. $P_2Q_3 \rightarrow 2P^{+3} + 3Q^{+2}$

$ksp = 2a\ 3a \qquad ksp = (2a)^2(3a)^3$

$ksp = 4\times a^5 \times 27 \qquad ksp = 108a^5$

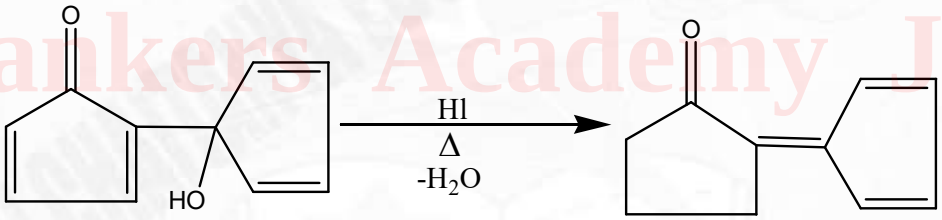
54. NO_2 is meta directing group. So the next ring will get – C1.

55. Order of Ea is $Cis > Trans$.

56. Finkel stein Reaction. Acetone is polar aprotic solvent.

57. -M groups increasing acidic nature of phenol.

58. Aldol Condensation.



59. (i) and (ii) are true, (i) and (iii) are true, (ii) and (iii) are true.

60. Ring attached with $-\overset{\overset{O}{\|}}{C}-$ has been deactivated by $-\underset{\underset{O}{\|}}{C}-$.

61. In Clemmenson Reduction $-\overset{\overset{O}{\|}}{C}-$ only reduced to $-CH_2-$ group.

62. H_2O

63. $-NH-\overset{\overset{O}{\|}}{C}-CH_3$ IS STRONGER +M group than $-CH_3$ group.

64. Statement-I & Statement-II both are true.

65. $[Fe(H_2O)_6]^{+2}$ has SP^3d^2 & $\left[Fe(CN)_6\right]^{-4}$ has d^2sp^3 hybridization.

66. $\Delta H^o_{hydration}$ $Cu^{+2} > Cu^{+1}$

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67.

$$ZnCl_2 \xrightarrow{AgNO_3 + dil\ HNO_3} \underset{white}{Agcl\downarrow} + \underset{(ag)}{Zn(NO_3)_2}$$

68. RLVP is colligative property.

69. When $k_1 = k_2$

$$10^{15}e^{-2000/T} = 10^{14}e^{-1000/T} = 10 = e^{1000/T} = 2.303\log 10 = \frac{1000}{T} = T = 434.2K$$

70. Phenylcation is unstable.

71. $\frac{(t_{1/2})_1}{(t_{1/2})_2} = \left(\frac{a_2}{a_1}\right)^{n-1}$ Where, n is order of reaction.

From the given data,

$$\frac{3.52}{1.92} = \left(\frac{13333}{6667}\right)^{n-1} \quad (a \propto \text{Initial Pressure}) = (2)^{n-1}$$

$$\log\frac{3.52}{1.92} = (n-1)\log 2 = 0.3010 \times (n-1)$$

$$0.2632 = 0.3010 \times (n-1)$$

$$n = 1.87 = 2\ .$$

72. $m = 3, l = 3, n = 4$

[For $n = 4$, members of waves will be 4].

73. $NO_2, SO_3, B_2O_3, Cl_2O_7, CrO_3$.

74. $Fe^{+2} \rightarrow Fe^{+2} + e^-$

$$2S^{-1} \rightarrow 2S^{+4} + 10e^-$$

Total number of electrons lost = $11e^-$

75.

$$\Delta G = \Delta G^o + RT\ In\ Q$$

$$\Rightarrow -nFE^o + 2.30RT\log\frac{\left[Cr^{3+}\right]}{\left[Ag^3\right]^3}$$

$$\Delta G = -3 \times 96500 \times 1.50 + 8.314298 \times 2.303\frac{(0.1)}{(0.1)^3}$$

$$\Rightarrow -422838.3J \ \ or\ -422.83kJ$$

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